CSE4062 – Introduction to Data Science and Analytics – Spring 2020 - Final Exam

RULES:

1- **This is an open-book exam; you may access class materials and slides during the exam, however you are NOT allowed to communicate and collaborate with other students and you are NOT allowed receive help from another person.**
2- **You need to write your answers to a white colored A4 size papers by your handwriting ONLY! Printer outputs will not be allowed.**
3- **You need to write your full name, your number, department, class code and date – time on the top of all of your answer papers.**
4- **In the first page, after writing the information in (3), please write the following statement and sign: "By signing this examination, I acknowledge my responsibility and commitment to the rules stated in the question paper and I affirm that I will not give or receive any unauthorized help on this exam, and that all work will be my own." After writing this statement and signing you can start answering questions.**
5- **When you finish scan all your answer papers in to a single pdf named "firstname_lastname.pdf" and upload the pdf to https://ues.marmara.edu.tr/**
6- **If you experience any technical problems, you can write a message to classroom.**

Points = { 1:20, 2:20, 3:20, 4:20, 5:20 ,6:25}

1-

a- What are the standard/basic steps we go through in a data science project. (Hint: One example is Jeff Hammerbacher's model for doing data science. Another example is Cross-industry standard process for data mining, known as CRISP-DM)
b- Explain briefly (in your own words!) your experience and your gains (what you learned) at each step in your term project.

2- Calculate following evaluation metrics by using the confusion matrix below for both "+" and "-" class. Show your calculations in detail!
(a) Accuracy
(b) Precision of + class
(c) Precision of - class
(d) Recall of + class
(e) Recall of - class

		Predicted	
		+	-
Actual	+	173	27
	-	44	456

3-
a- Define cluster concept clearly (if possible, mathematically) and based on this define clustering analysis. How does clustering differ from classification?
b- For the k-means algorithm, how can we experimentally find best k value? How did you defined k value in your term project.

4- In data visualization, which methods you use to see if
a- a nominal variable such as age has right skewed distribution (draw an example)
b- two numeric variables has negative correlation (draw an example)
c- three variables such as age, gender and salary has interesting correlations.
d- Which method(s) will you use if you want to visualize high-dimensional continuous data to see the clustering of points in 2D space?

5- Please provide the following information regarding your project **in your own words!**(This question will be evaluated based on your GitHub project account by the time of the final exam).
a- Project group number and your project topic
b- List attribute/feature selection methods used (if any) in your project
c- List machine learning algorithms used in your project. Indicate if the algorithm is supervised or unsupervised along with any algorithms parameters you tried (such as kernel or C value in SVM)
d- List evaluation metrics used in your project
e- Summarize your results, which algorithms worked best on your dataset, under which conditions and by which evaluation metrics?

6- Consider the following training data about playing tennis.
a- Convert all attributes (except Class) to numeric attributes use one-hot encoding and show this new dataset in a table.
b- Classify (predict class label of) the following instance X = <sunny, cool, high, true> using k-NN algorithm (k=1) and Manhattan Distance. Please remember distance is the opposite of similarity. Show your calculations in detail! Explain the decision.

Outlook	Temperature	Humidity	Windy	Class
sunny	hot	high	false	N
sunny	hot	high	true	N
overcast	hot	high	false	P
rain	mild	high	false	P
rain	cool	normal	false	P
rain	cool	normal	true	N
overcast	cool	normal	true	P
sunny	mild	high	false	N
sunny	cool	normal	false	P
rain	mild	normal	false	P